Algorithm Design Examination

Reconstructed Questions

Scoring. Problems 1 and 6 are worth 2 points each; each of the other six problems is worth 3 points. The total raw score is 22 points. If the raw score is s, the final score is $\min\{s, 20\}$.

Problem 1: Asymptotic Growth Rates (2 points)

Order all six functions below by increasing asymptotic growth rate:

(1) $(\sqrt{n})^{\log\log n}$;
(2) $2^{2^{\log^* n}}$;
(3) $\binom{n}{\lfloor \log\log n \rfloor}$;
(4) $n^{(\log\log\log n)^2}$;
(5) $2^{(\log\log n)^2}$;
(6) $2^{2^{\log\log n}}$.

Here $\log^* n$ is the number of times the logarithm must be applied until the result is at most a fixed constant. The specific stopping constant need not be specified.

Problem 2: Prim's Algorithm (3 points)

Let $G = (V, E)$ be a simple, connected, weighted, undirected graph with $n = |V|$ and $m = |E|$.

(1) Assume that all edge weights are positive. Write pseudocode for Prim's algorithm and analyze its running time. You may use a priority queue directly without implementing its internal operations.
(2) Prove or disprove that the correctness of Prim's algorithm requires positive edge weights. Discuss its correctness when the weights are positive, negative, or a mixture of both.

Problem 3: Subsequence Testing (3 points)

You are given two arrays S' and S of lengths m and n, respectively. Determine whether S' is a subsequence of S.

An array S' is a *subsequence* of S if deleting zero or more elements from S, while preserving the relative order of the remaining elements, produces S'. Equivalently, there exist indices

$$1 \le i_1 < i_2 < \cdots < i_m \le n, \qquad S'[j] = S[i_j] \quad (1 \le j \le m).$$

Design an $O(n + m)$-time algorithm and justify its correctness.

Problem 4: Allocating Time to Homework (3 points)

There are n homework assignments and a total time budget of H. For each assignment i, you are given a function

$$f_i : \{0, 1, \ldots, H\} \longrightarrow \{1, 2, \ldots, g\},$$

where $f_i(h)$ is the benefit obtained by spending h time units on assignment i. All function values for integer $h = 0, \ldots, H$ are explicitly provided; reading the entire input takes $O(nH)$ time.

Each function is nondecreasing:

$$0 \le h < h' \le H \quad \Longrightarrow \quad f_i(h') \ge f_i(h).$$

Choose nonnegative integers $h_1, \ldots, h_n$ with total at most H. Maximize the total benefit $\sum_{i=1}^{n} f_i(h_i)$. You do not have to use all the available time.

Give a dynamic programming algorithm whose running time is polynomial in n, H, g. Your algorithm must return both the optimal value and a time allocation attaining it.

Problem 5: Path Selection (3 points)

You are given an undirected graph $G = (V, E)$ and v users. Each user j is associated with an explicitly given path P_j in G, for $1 \le j \le v$. Two paths intersect if they share a vertex.

Given an integer k, determine whether it is possible to select k users whose paths are pairwise vertex-disjoint. Formally, determine whether there exists

$$J \subseteq \{1, \ldots, v\}, \quad |J| = k, \qquad V(P_a) \cap V(P_b) = \varnothing \quad (\forall a, b \in J,\ a \ne b).$$

Prove that this decision problem is NP-complete.

Problem 6: Matroids and Pseudoforests (2 points)

Let E be a finite ground set and let $\mathcal{I} \subseteq 2^E$. The pair $(E, \mathcal{I})$ is a *matroid* if:

(1) $\varnothing \in \mathcal{I}$.
(2) If $A \subseteq B$ and $B \in \mathcal{I}$, then $A \in \mathcal{I}$.
(3) If $A, B \in \mathcal{I}$ and $|A| < |B|$, then there exists $e \in B \setminus A$ such that $A \cup \{e\} \in \mathcal{I}$.

Given an undirected graph $G = (V, E)$, an edge set $F \subseteq E$ is a *pseudoforest* if every connected component of (V, F) contains at most one cycle. Define

$$\mathcal{I} = \{F \subseteq E : (V, F) \text{ is a pseudoforest}\}.$$

Prove or disprove that $(E, \mathcal{I})$ is a matroid.

Problem 7: NAE and the NP-Completeness of Max-Cut (3 points)

A *literal* is a Boolean variable or its negation. In 3SAT, the input is a conjunction of clauses, each containing three literals, and the question is whether a truth assignment satisfies every clause.

NAE stands for *Not All Equal*. A clause satisfies the NAE condition if it contains at least one true literal and at least one false literal. Formally, for $b_1, \ldots, b_r \in \{0, 1\}$,

$$\mathrm{NAE}(b_1, \ldots, b_r) = \begin{cases} 1, & \text{if both 0 and 1 occur among } b_1, \ldots, b_r, \\ 0, & \text{if all the values are equal.} \end{cases}$$

For example, $\mathrm{NAE}(1, 0, 1) = 1$, whereas $\mathrm{NAE}(1, 1, 1) = \mathrm{NAE}(0, 0, 0) = 0$.

In NAE-r, the input consists of clauses with r literals each. **Repeated literals within a clause are not allowed.** The question is whether there exists a truth assignment under which *every* clause satisfies the NAE condition.

(1) Prove that $3SAT \leq_p NAE\text{-}4$.
(2) Prove that $NAE\text{-}4 \leq_p NAE\text{-}3$.
(3) By reducing from NAE-3, prove that Max-Cut is NP-complete.

Here $\leq_p$ denotes polynomial-time reduction. The decision version of Max-Cut takes an undirected graph $G = (V, E)$ with edge weights w_e, together with a threshold q, and asks whether there exists $S \subseteq V$ such that

$$\sum_{e \in \delta(S)} w_e \geq q,$$

where $\delta(S)$ is the set of edges with one endpoint in S and the other in $V \setminus S$.

Problem 8: Budgeted Independent Set (3 points)

You are given an undirected graph $G = (V, E)$ with $|V| = N$, together with a tree decomposition

$$(T = (I, E_T), \{B_t\}_{t \in I})$$

of width at most k, where k is a constant. Thus every bag satisfies $|B_t| \leq k + 1$, and $|I|$ is the number of bags.

Each vertex v has a nonnegative cost $\mathrm{cost}(v)$ and a nonnegative profit $\mathrm{profit}(v)$. You are also given a budget B.

Select a vertex set $S \subseteq V$ such that:

(1) S is an independent set in G: no two selected vertices are adjacent.
(2) $\sum_{v \in S} \mathrm{cost}(v) \leq B$.

The objective is to maximize

$$\sum_{v \in S} \mathrm{profit}(v).$$

Design an algorithm whose running time is polynomial in N, $|I|$, and B.